

ADVANCED

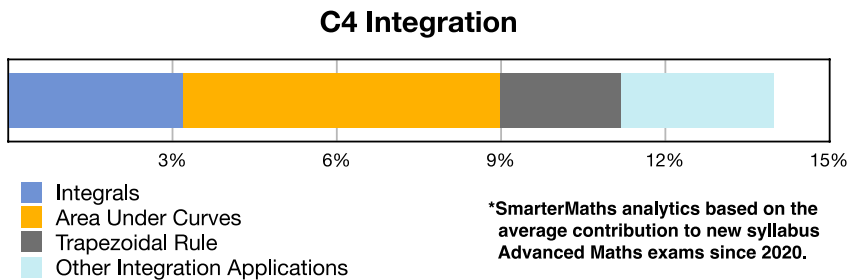
Stage 6

Calculus (Adv), C4 Integration (Adv)

Trapezoidal Rule (Y12)

Teacher: SHS Maths

Exam Equivalent Time: 90 minutes (based on allocation of 1.5 minutes per mark)



HISTORICAL CONTRIBUTION

- C4 Integration* has contributed a substantial 14.0% to new syllabus Advanced exams since it was introduced in 2020.
- This topic has been split into four sub-topics for analysis purposes: 1-*Integrals* (3.2%), 2-*Areas Under Curves* (5.8%), 3-*Trapezoidal Rule* (2.2%) and 4-*Other Integration Examples* (2.8%).
- This analysis looks at *Trapezoidal Rule*.

HSC ANALYSIS - What to expect and common pitfalls

- Trapezoidal Rule* (2.2%) is the surviving *area approximation method* in the new Advanced course, with Simpson's Rule no longer examinable.
- Trapezoidal Rule* has been examined in 3 of the last 4 new syllabus exams (most recently in 2024), including two separate questions in 2022 (review 2022 Adv 29 - a cracking question).
- Note that while the standard formula and calculations are generally well answered, the follow up questions have produced sub-50% mean marks 4 times in the last decade.
- Where possible, past Simpson Rule approximation questions have been rewritten as Trapezoidal Rule questions to help address the problem areas identified above.

Questions

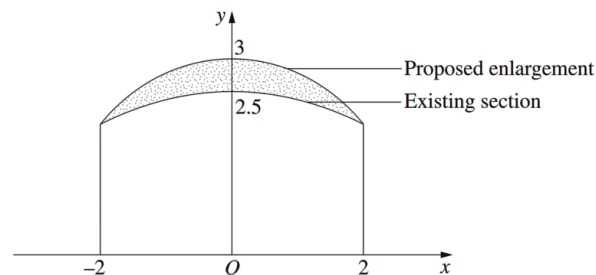
1. Calculus, 2ADV C4 2015 HSC 5 MC

Using the trapezoidal rule with 4 subintervals, which expression gives the approximate area under the curve $y = xe^x$ between $x = 1$ and $x = 3$?

- A. $\frac{1}{4} (e^1 + 6e^{1.5} + 4e^2 + 10e^{2.5} + 3e^3)$
- B. $\frac{1}{4} (e^1 + 3e^{1.5} + 4e^2 + 5e^{2.5} + 3e^3)$
- C. $\frac{1}{2} (e^1 + 6e^{1.5} + 4e^2 + 10e^{2.5} + 3e^3)$
- D. $\frac{1}{2} (e^1 + 3e^{1.5} + 4e^2 + 5e^{2.5} + 3e^3)$

2. Calculus, 2ADV C4 2016* HSC 14a

The diagram shows the cross-section of a tunnel and a proposed enlargement.



The heights, in metres, of the existing section at 1 metre intervals are shown in Table A.

Table A: Existing heights					
x	-2	-1	0	1	2
y	2	2.38	2.5	2.38	2

The heights, in metres, of the proposed enlargement are shown in Table B.

Table B: Proposed heights					
x	-2	-1	0	1	2
y	2	2.78	3	2.78	2

Use the Trapezoidal rule with the measurements given to calculate the approximate increase in area. (3 marks)

3. Calculus, 2ADV C4 2020 HSC 20

Kenzo is driving his car along a road while his friend records the velocity of the car, $v(t)$, in km/h every minute over a 5-minute period. The table gives the velocity $v(t)$ at time t hours.

t	0	$\frac{1}{60}$	$\frac{2}{60}$	$\frac{3}{60}$	$\frac{4}{60}$	$\frac{5}{60}$
$v(t)$	60	55	65	68	70	67

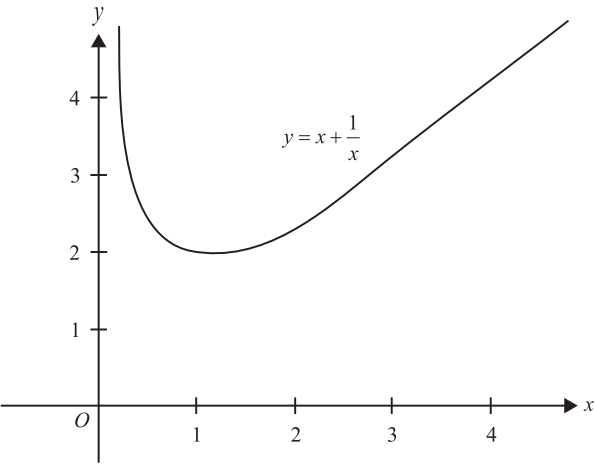
The distance covered by the car over the 5-minute period is given by

$$\int_0^{\frac{5}{60}} v(t) \, dt.$$

Use the trapezoidal rule and the velocity at each of the six time values to find the approximate distance in kilometres the car has travelled in the 5-minute period. Give your answer correct to one decimal place. **(2 marks)**

4. Calculus, 2ADV C4 2022 MET1 4

The graph of $y = x + \frac{1}{x}$ is shown over part of its domain.



Use two trapeziums of equal width to approximate the area between the curve, the x -axis and the lines $x = 1$ and $x = 3$. **(2 marks)**

5. Calculus, 2ADV C4 2006* HSC 10a

Use the Trapezoidal rule with three function values to find an approximation to the value of

$$\int_{0.5}^{1.5} (\log_e x)^3 \, dx.$$

Give your answer correct to three decimal places. **(2 marks)**

6. Calculus, 2ADV C4 2011* HSC 5c

The table gives the speed v of a jogger at time t in minutes over a 20-minute period. The speed v is measured in metres per minute, in intervals of 5 minutes.

t	0	5	10	15	20
v	173	81	127	195	168

The distance covered by the jogger over the 20-minute period is given by $\int_0^{20} v \, dt$.

Use the Trapezoidal rule and the speed at each of the five time values to find the approximate distance the jogger covers in the 20-minute period. **(3 marks)**

7. Calculus, 2ADV C4 2005* HSC 6a

Five values of the function $f(x)$ are shown in the table.

x	0	5	10	15	20
$f(x)$	15	25	22	18	10

Use the Trapezoidal rule with the five values given in the table to estimate

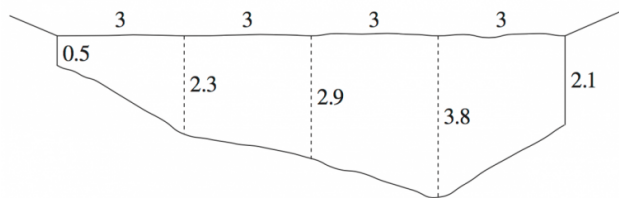
$$\int_0^{20} f(x) \, dx. \quad \textbf{(3 marks)}$$

8. Calculus, 2ADV C4 2022 HSC 13

Use two applications of the trapezoidal rule to find an approximate value of $\int_0^2 \sqrt{1+x^2} \, dx$. Give your answer correct to 2 decimal places. **(2 marks)**

9. Calculus, 2ADV C4 2012* HSC 12d

At a certain location a river is 12 metres wide. At this location the depth of the river, in metres, has been measured at 3 metre intervals. The cross-section is shown below.



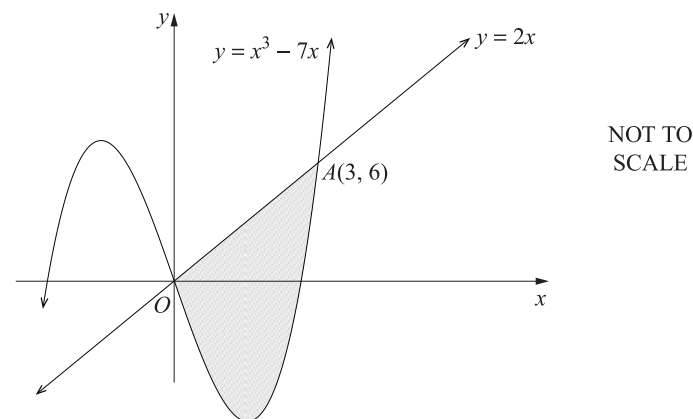
- Use the Trapezoidal rule with the five depth measurements to calculate the approximate area of the cross-section. **(3 marks)**
- The river flows at 0.4 metres per second.
Calculate the approximate volume of water flowing through the cross-section in 10 seconds. **(1 mark)**

10. Calculus, 2ADV C4 2017* HSC 14b

- Find the exact value of $\int_0^{\frac{\pi}{3}} \cos x \, dx$. **(1 mark)**
- Using the Trapezoidal rule with three function values, find an approximation to the integral $\int_0^{\frac{\pi}{3}} \cos x \, dx$, leaving your answer in terms of π and $\sqrt{3}$. **(2 marks)**
- Using parts (i) and (ii), show that $\pi \approx \frac{12\sqrt{3}}{3 + 2\sqrt{3}}$. **(1 mark)**

11. Calculus, 2ADV C4 2018* HSC 15c

The shaded region is enclosed by the curve $y = x^3 - 7x$ and the line $y = 2x$, as shown in the diagram. The line $y = 2x$ meets the curve $y = x^3 - 7x$ at $O(0, 0)$ and $A(3, 6)$. Do NOT prove this.

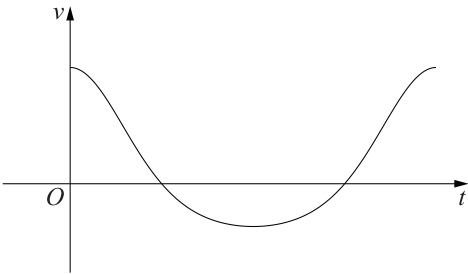


- Use integration to find the area of the shaded region. **(2 marks)**
- Use the Trapezoidal rule and four function values to approximate the area of the shaded region. **(2 marks)**
The point P is chosen on the curve $y = x^3 - 7x$ so that the tangent at P is parallel to the line $y = 2x$ and the x -coordinate of P is positive
- Show that the coordinates of P are $(\sqrt{3}, -4\sqrt{3})$. **(2 marks)**
- Using the perpendicular distance formula $\frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$, find the area of $\triangle OAP$. **(2 marks)**

12. Calculus, 2ADV C4 2019 HSC 16b

A particle moves in a straight line, starting at the origin. Its velocity, $v \text{ ms}^{-1}$, is given by $v = e^{\cos t} - 1$, where t is in seconds.

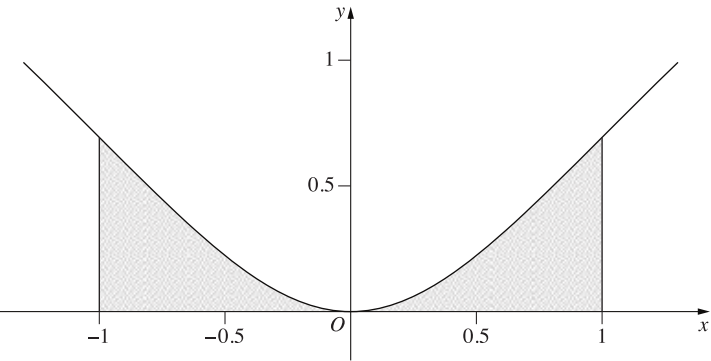
The diagram shows the graph of the velocity against time.



Using the Trapezoidal Rule with three function values, estimate the position of the particle when it first comes to rest. Give your answer correct to two decimal places. (3 marks)

13. Calculus, 2ADV C4 2024 HSC 22

The graph of the function $f(x) = \ln(1 + x^2)$ is shown.



- a. Prove that $f(x)$ is concave up for $-1 < x < 1$. (3 marks)
- b. A table of function values, correct to 4 decimal places, for some x values is provided.

x	0	0.25	0.5	0.75	1
$\ln(1 + x^2)$	0	0.0606	0.2231	0.4463	0.6931

Using the function values provided and the trapezoidal rule, estimate the shaded area in the diagram. (2 marks)

- c. Is the answer to part (b) an overestimate or underestimate? Give a reason for your answer. (1 mark)

14. Calculus, 2ADV C4 2004* HSC 10a

- i. Use the Trapezoidal rule with 3 function values to find an approximation to the area under the curve $y = \frac{1}{x}$ between $x = a$ and $x = 3a$, where a is positive. (2 marks)
- ii. Using the result in part (i), show that $\ln 3 \div \frac{7}{6}$. (1 mark)

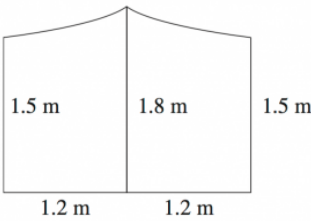
15. Calculus, 2ADV C4 2014* HSC 16a

Use the Trapezoidal rule with five function values to show that

$$\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \sec x \, dx \approx \frac{\pi}{6} \left(3 + \frac{4}{\sqrt{3}} \right). \quad (3 \text{ marks})$$

16. Calculus, 2ADV C4 2013* HSC 15a

The diagram shows the front of a tent supported by three vertical poles. The poles are 1.2 m apart. The height of each outer pole is 1.5 m, and the height of the middle pole is 1.8 m. The roof hangs between the poles.

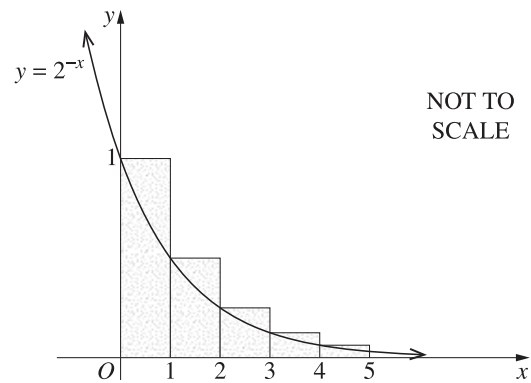


The front of the tent has area $A \text{ m}^2$.

- i. Use the trapezoidal rule to estimate A . (1 mark)
- ii. Does the Trapezoidal rule give a higher or lower estimate of the actual area? Justify your answer. (1 mark)

17. Calculus, 2ADV C4 2022 HSC 29

- a. The diagram shows the graph of $y = 2^{-x}$. Also shown on the diagram are the first 5 of an infinite number of rectangular strips of width 1 unit and height $y = 2^{-x}$ for non-negative integer values of x . For example, the second rectangle shown has width 1 and height $\frac{1}{2}$.



The sum of the areas of the rectangles forms a geometric series.

Show that the limiting sum of this series is 2. (1 mark)

- b. Show that $\int_0^4 2^{-x} dx = \frac{15}{16\ln 2}$. (2 marks)
- c. Use parts (a) and (b) to show that $e^{15} < 2^{32}$. (2 marks)

18. Calculus, 2ADV C4 2010 HSC 3b

- i. Sketch the curve $y = \ln x$. (1 mark)
- ii. Use the trapezoidal rule with 3 function values to find an approximation to $\int_1^3 \ln x dx$ (2 marks)
- iii. State whether the approximation found in part (ii) is greater than or less than the exact value of $\int_1^3 \ln x dx$. Justify your answer. (1 mark)

Worked Solutions

1. Calculus, 2ADV C4 2015 HSC 5 MC

$y = xe^x$

x	1	1.5	2	2.5	3
y	e^1	$1.5e^{1.5}$	$2e^2$	$2.5e^{2.5}$	$3e^3$
	y_0	y_1	y_2	y_3	y_4

$$A \approx \frac{h}{2} [y_0 + 2y_1 + 2y_2 + 2y_3 + y_4]$$
$$\approx \frac{1}{4} [e^1 + 3e^{1.5} + 4e^2 + 5e^{2.5} + 3e^3]$$
$$\Rightarrow B$$

2. Calculus, 2ADV C4 2016* HSC 14a

Consider the shaded area distances:

x	-2	-1	0	1	2
height	0	0.4	0.5	0.4	0
weight	1	2	2	2	1

$$A \approx \frac{1}{2} [0 + 2(0.4 + 0.5 + 0.4) + 0]$$
$$\approx \frac{1}{2} (2.6)$$
$$\approx 1.3 \text{ m}^2$$

3. Calculus, 2ADV C4 2020 HSC 20

t	0	$\frac{1}{60}$	$\frac{2}{60}$	$\frac{3}{60}$	$\frac{4}{60}$	$\frac{5}{60}$
$v(t)$	60	55	65	68	70	67
weight	1	2	2	2	2	1

$$\begin{aligned}\int_0^{\frac{5}{60}} v(t) \, dt &\approx \frac{1}{2} \times \frac{1}{60} [60 + 2(55 + 65 + 68 + 70) + 67] \\ &\approx \frac{1}{120} (643) \\ &\approx 5.358\dots \\ &\approx 5.4 \text{ km}\end{aligned}$$

4. Calculus, 2ADV C4 2022 MET1 4

Trapezium rule approximation (see formula sheet):

x	1	2	3
$f(x)$	$1 + 1 = 2$	$2 + \frac{1}{2} = \frac{5}{2}$	$3 + \frac{1}{3} = \frac{10}{3}$

$$\begin{aligned}\text{Area} &\approx \frac{3-1}{2 \times 2} \left[2 + 2 \times \frac{5}{2} + \frac{10}{3} \right] \\ &\approx \frac{1}{2} \left[\frac{6}{3} + \frac{15}{3} + \frac{10}{3} \right] \\ &\approx 5\frac{1}{6}\end{aligned}$$

5. Calculus, 2ADV C4 2006* HSC 10a

Let $f(x) = (\log_e x)^3$

x	0.5	1	1.5
$f(x)$	-0.3330...	0	0.0666...
weight	1	2	1

$$\begin{aligned}\int_{0.5}^{1.5} (\log_e x)^3 \, dx &\approx \frac{0.5}{2} [-0.3330\dots + 2(0) + 0.0666\dots] \\ &\approx 0.25(-0.2663\dots) \\ &\approx -0.06659\dots \\ &\approx -0.067 \quad (\text{to 3 d.p.})\end{aligned}$$

6. Calculus, 2ADV C4 2011* HSC 5c

t	0	5	10	15	20
v	173	81	127	195	168
weight	1	2	2	2	1

$$\begin{aligned}\int_0^{20} v \, dt &\approx \frac{5}{2} [173 + 2(81 + 127 + 195) + 168] \\ &\approx \frac{5}{2} (1147) \\ &\approx 2867.5 \text{ metres}\end{aligned}$$

7. Calculus, 2ADV C4 2005* HSC 6a

x	0	5	10	15	20
$f(x)$	15	25	22	18	10
weight	1	2	2	2	1

$$\begin{aligned}\therefore \int_0^{20} f(x) \, dx &\approx \frac{5}{2} [15 + 2(25 + 22 + 18) + 10] \\ &\approx \frac{5}{2} (155) \\ &\approx 387.5\end{aligned}$$

8. Calculus, 2ADV C4 2022 HSC 13

$$A \approx \frac{h}{2}(y_0 + 2y_1 + y_2)$$
$$\approx \frac{1}{2}\left(1 + 2 \times \sqrt{2} + \sqrt{5}\right)$$
$$\approx 3.03 \text{ (to 2 d.p.)}$$

Mean mark 59%.
COMMENT: A surprisingly low mean mark warrants attention.

9. Calculus, 2ADV C4 2012* HSC 12d

i.

x	0	3	6	9	12
height	0.5	2.3	2.9	3.8	2.1
weight	1	2	2	2	1

$$A \approx \frac{3}{2}[0.5 + 2(2.3 + 2.9 + 3.8) + 2.1]$$
$$\approx \frac{3}{2}(20.6)$$
$$\approx 30.9 \text{ m}^2$$

ii. Distance water flows = 0.4×10
= 4 metres

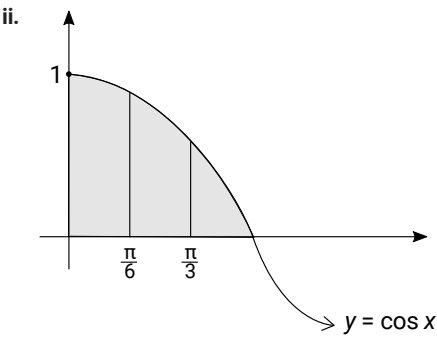
♦ Mean mark 49%.

Volume flow in 10 seconds $\approx 4 \times 30.9$
 $\approx 123.6 \text{ m}^3$

10. Calculus, 2ADV C4 2017* HSC 14b

i. $\int_0^{\frac{\pi}{3}} \cos x \, dx = [\sin x]_0^{\frac{\pi}{3}}$

$$= \sin \frac{\pi}{3} - 0$$
$$= \frac{\sqrt{3}}{2}$$



x	0	$\frac{\pi}{6}$	$\frac{\pi}{3}$
height	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$
weight	1	2	1

$$\int_0^{\frac{\pi}{3}} \cos x \, dx \approx \frac{1}{2} \times \frac{\pi}{6} \left[1 + 2 \left(\frac{\sqrt{3}}{2} \right) + \frac{1}{2} \right]$$
$$\approx \frac{\pi}{12} \left(\frac{3 + 2\sqrt{3}}{2} \right)$$
$$\approx \frac{(3 + 2\sqrt{3})\pi}{24}$$

(iii) $\frac{(3 + 2\sqrt{3})\pi}{24} \approx \frac{\sqrt{3}}{2}$

$$\therefore \pi \approx \frac{24\sqrt{3}}{2(3 + 2\sqrt{3})}$$
$$\approx \frac{12\sqrt{3}}{3 + 2\sqrt{3}}$$

♦ Mean mark part (iii) 49%.

11. Calculus, 2ADV C4 2018* HSC 15c

i.
$$\begin{aligned} \text{Area} &= \int_0^3 2x - (x^3 - 7x) \, dx \\ &= \int_0^3 9x - x^3 \, dx \\ &= \left[\frac{9}{2}x^2 - \frac{1}{4}x^4 \right]_0^3 \\ &= \left[\left(\frac{9}{2} \times 3^2 - \frac{1}{4} \times 3^4 \right) - 0 \right] \\ &= \frac{81}{2} - \frac{81}{4} \\ &= \frac{81}{4} \text{ units}^2 \end{aligned}$$

ii. $f(x) = 9x - x^3$

x	0	1	2	3
$f(x)$	0	8	10	0
weight	1	2	2	1

$$\begin{aligned} \text{Area} &\approx \frac{1}{2}[0 + 2(8 + 10) + 0] \\ &\approx \frac{1}{2}(36) \\ &\approx 18 \text{ u}^2 \end{aligned}$$

iii. $y = x^3 - 7x$

$$\frac{dy}{dx} = 3x^2 - 7$$

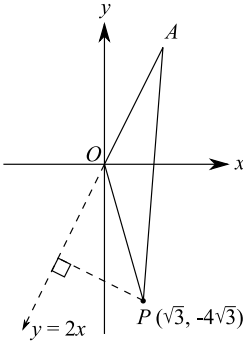
Find x such that $\frac{dy}{dx} = 2$

$$\begin{aligned} 3x^2 - 7 &= 2 \\ 3x^2 &= 9 \\ x^2 &= 3 \\ x &= \sqrt{3} \quad (x > 0) \end{aligned}$$

$$\begin{aligned} y &= (\sqrt{3})^3 - 7\sqrt{3} \\ &= 3\sqrt{3} - 7\sqrt{3} \\ &= -4\sqrt{3} \end{aligned}$$

$\therefore P$ has coordinates $(\sqrt{3}, -4\sqrt{3})$

iv.



$$\begin{aligned} \text{dist } OA &= \sqrt{(3-0)^2 + (6-0)^2} \\ &= \sqrt{45} \\ &= 3\sqrt{5} \end{aligned}$$

Find $\perp$ distance of P from OA

$$P(\sqrt{3}, -4\sqrt{3}), \quad 2x - y = 0$$

$$\begin{aligned} \perp \text{ dist} &= \left| \frac{2\sqrt{3} + 4\sqrt{3}}{\sqrt{3} + 2} \right| \\ &= \frac{6\sqrt{3}}{\sqrt{5}} \end{aligned}$$

$$\begin{aligned} \therefore \text{Area} &= \frac{1}{2} \times 3\sqrt{5} \times \frac{6\sqrt{3}}{\sqrt{5}} \\ &= 9\sqrt{3} \text{ units}^2 \end{aligned}$$

12. Calculus, 2ADV C4 2019 HSC 16b

$v = e^{\cos t} - 1$
Find t when $v = 0$:
 $e^{\cos t} = 1$
 $\cos t = 0$
 $t = \frac{\pi}{2}$

t	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$
v	$e-1$	$e^{\frac{1}{\sqrt{2}}}-1$	0
	v_0	v_1	v_2

$A \approx \frac{h}{2}(v_0 + 2v_1 + v_2)$
 $\approx \frac{\pi}{8}\left[e-1 + 2\left(e^{\frac{1}{\sqrt{2}}}-1\right) + 0\right]$
 $\approx \frac{\pi}{8}(3.774\dots)$
 $\approx 1.482\dots$
 $\approx 1.48 \text{ (2 d.p.)}$

$\therefore$ The particle will be 1.48 metres to the right when it comes to rest.

♦♦ Mean mark 30%.

13. Calculus, 2ADV C4 2024 HSC 22

a. $f(x) = \ln(1 + x^2)$
 $f'(x) = \frac{2x}{1 + x^2}$
 $f''(x) = \frac{2(1 + x^2) - 2x(2x)}{(1 + x^2)^2}$
 $= \frac{2 + 2x^2 - 4x^2}{(1 + x^2)^2}$
 $= \frac{2(1 - x^2)}{(1 + x^2)^2}$

In domain $x \in (-1, 1) \Rightarrow 1 - x^2 > 0$
 $(1 + x^2)^2 > 0$ for all x
 $\Rightarrow f''(x) > 0$ for $x \in (-1, 1)$
 $\therefore f(x)$ is concave up for $x \in (-1, 1)$

b. Total shaded area

$\approx 2 \times \frac{h}{2}[y_0 + 2(y_1 + y_2 + y_3) + y_4]$
 $\approx 2 \times \frac{0.25}{2}[0 + 2(0.0606 + 0.2231 + 0.4463) + 0.6931]$
 ≈ 0.538275
 $\approx 0.5383 \text{ (4 d.p.)}$

c. Trapezoidal rule assumes straight lines join points in the table.

Since the graph is concave up in the given domain, the trapezoidal rule will overestimate the area.

♦ Mean mark (a) 47%.

♦ Mean mark (b) 50%.

♦ Mean mark (c) 49%.

14. Calculus, 2ADV C4 2004* HSC 10a

i.

x	a	$2a$	$3a$
y	$\frac{1}{a}$	$\frac{1}{2a}$	$\frac{1}{3a}$
weight	1	2	1

$$\begin{aligned} A &\approx \frac{a}{2} \left[\frac{1}{a} + 2 \left(\frac{1}{2a} \right) + \frac{1}{3a} \right] \\ &\approx \frac{a}{2} \left(\frac{7}{3a} \right) \\ &\approx \frac{7}{6} \end{aligned}$$

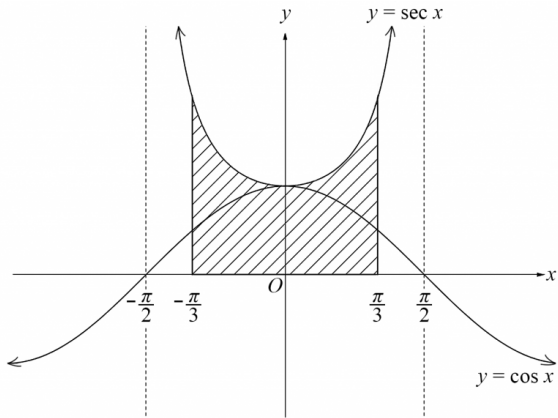
ii. Area under the curve $y = \frac{1}{x}$

$$\begin{aligned} &= \int_a^{3a} \frac{1}{x} dx \\ &= [\ln x]_a^{3a} \\ &= \ln 3a - \ln a \\ &= \ln \frac{3a}{a} \\ &= \ln 3 \end{aligned}$$

Trapezoidal rule in part (i) found the approximate value of the same area.

$$\therefore \ln 3 \div \frac{7}{6}.$$

15. Calculus, 2ADV C4 2014* HSC 16a



◆ Mean mark below 50%. BE CAREFUL!

x	$-\frac{\pi}{3}$	$-\frac{\pi}{6}$	0	$\frac{\pi}{6}$	$\frac{\pi}{3}$
y	2	$\frac{2}{\sqrt{3}}$	1	$\frac{2}{\sqrt{3}}$	2
weight	1	2	2	2	1

$$\begin{aligned} A &\approx \frac{h}{2} [y_0 + 2(y_1 + y_2 + y_3) + y_5] \\ &\approx \frac{\pi}{12} \left[2 + 2 \left(\frac{2}{\sqrt{3}} + 1 + \frac{2}{\sqrt{3}} \right) + 2 \right] \\ &\approx \frac{\pi}{12} \left[6 + \frac{8}{\sqrt{3}} \right] \\ &\approx \frac{\pi}{6} \left(3 + \frac{4}{\sqrt{3}} \right) \text{ u}^2 \dots \text{ as required} \end{aligned}$$

16. Calculus, 2ADV C4 2013* HSC 15a

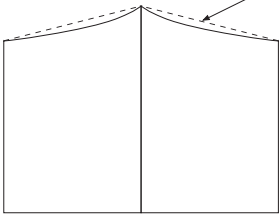
i. $A \approx \frac{h}{2} [y_0 + 2y_1 + y_2]$

$\approx \frac{1.2}{2} [1.5 + (2 \times 1.8) + 1.5]$

$\approx 0.6 [6.6]$

$\approx 3.96 \text{ m}^2$

ii. Trapezoidal rule



Since the tent roof is concave up, the Trapezoidal rule uses straight lines and will estimate a higher area.

17. Calculus, 2ADV C4 2022 HSC 29

a. Consider the rectangle heights:

$2^0 = 1, \ 2^{-1} = \frac{1}{2}, \ 2^{-2} = \frac{1}{4}, \ 2^{-3} = \frac{1}{8}, \dots$

$\Rightarrow \text{Rectangle Areas} = 1, \ \frac{1}{2}, \ \frac{1}{4}, \ \frac{1}{8}, \dots$

$a = 1, \ r = \frac{1}{2}$

$S_\infty = \frac{a}{1 - r} = \frac{1}{1 - \frac{1}{2}} = 2 \dots \text{as required}$

b. Show $\int_0^4 2^{-x} \, dx = \frac{15}{16\ln 2}$

$\int_0^4 2^{-x} \, dx = \frac{-1}{\ln 2} [2^{-x}]_0^4$

$= \frac{-1}{\ln 2} \left(\frac{1}{16} - 1 \right)$

$= \frac{1}{\ln 2} - \frac{1}{16\ln 2}$

$= \frac{16 - 1}{16\ln 2}$

$= \frac{15}{16\ln 2} \dots \text{as required}$

Mean mark (b) 56%.

c. Show $e^{15} < 2^{32}$

Area under curve < Sum of rectangle areas

$\frac{15}{16\ln 2} < 2$

$15 < 32\ln 2$

$\frac{15}{32} < \ln 2$

$e^{\frac{15}{32}} < e^{\ln 2}$

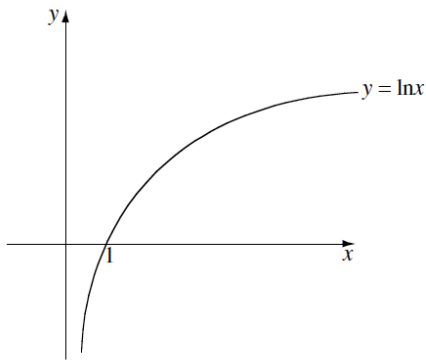
$\sqrt[32]{e^{15}} < 2$

$e^{15} < 2^{32} \dots \text{as required}$

♦♦♦ Mean mark (c) 9%.

18. Calculus, 2ADV C4 2010 HSC 3b

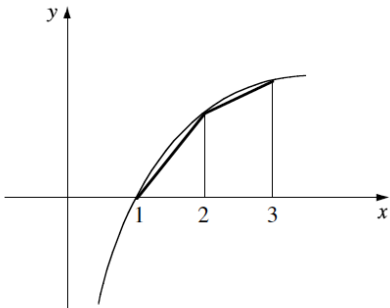
i.



ii.
$$\begin{aligned} \text{Area} &\approx \frac{h}{2} [f(1) + 2 \times f(2) + f(3)] \\ &\approx \frac{1}{2} [0 + 2\ln 2 + \ln 3] \\ &\approx \frac{1}{2} [\ln(2^2 \times 3)] \\ &\approx \frac{1}{2} \ln 12 \\ &\approx 1.24 \text{ u}^2 \quad (\text{to 2 d.p.}) \end{aligned}$$

MARKER'S COMMENT: Many students failed to illustrate important features in their graph such as the concavity, x -axis intercept and y -axis asymptote (this can be explicitly stated or made graphically clear).

iii.



The approximation is less because the sides of the trapezia lie below the concave down curve (see diagram).

◆◆ Mean mark 12%.
MARKER'S COMMENT: Best responses commented on concavity and that the trapezia lay beneath the curve. Diagrams featured in the best responses.